RESEARCH PLAN

IngreGuide: A Web-Based Personalized Food Ingredient Information System Using Barcode Scanning

Proponents

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Rationale

Consumers often find packaged-food ingredient labels difficult to understand because information is presented in dense lists, small text, and unfamiliar terminology. Older adults and consumers with specific dietary requirements may have greater difficulty interpreting ingredient information, particularly when several ingredients are presented using technical names or abbreviations (Lin & Wang, 2022; Hall et al., 2022). Studies also indicate that consumers' food-label knowledge and practical understanding vary across populations, affecting how effectively they use labels when making food choices (Icer & Karadağ, 2023; Islam et al., 2024). These problems can prevent consumers from identifying relevant ingredients, comparing products, or determining whether a product is appropriate for their dietary needs.

Technical and unfamiliar ingredient terminology creates an additional barrier to food-label comprehension. Scientific names, additive names, and chemical-sounding terms may be difficult for consumers to interpret, causing uncertainty about an ingredient's purpose, safety, or nutritional significance. Research shows that consumers' attitudes toward additives are influenced by their level of knowledge and their perceptions of product safety and transparency (Osaili et al., 2023). Unfamiliar ingredients may also reduce perceptions of naturalness or environmental friendliness, demonstrating that ingredient terminology can influence product judgments beyond simple comprehension (Li et al., 2024). Therefore, consumers may benefit from a system that translates complex ingredient names into brief, understandable explanations.

Checking allergens in packaged foods is important especially for people who have food allergies and need to avoid certain ingredients. However, allergen warnings are not always easy to understand and less than 40% of the participants in one study found precautionary allergen information easy to understand (Holleman et al., 2021). In the Philippines, only 19.6% of adults said they always read food labels, while 45.9% said they do not read them (Javier et al., 2024). This can make it harder for consumers to notice ingredients that may be unsafe for them, therefore having a system that makes ingredient and allergen information easier to find which may help users check food products more carefully.

People do not all have the same food needs since some may have allergies, dietary restrictions, or personal food preferences. Because of this, general information about a product may not be equally useful to everyone. One study found

that 25.7% of 1,626 participants used at least one food barcode scanner app to check food quality and other food information (Hanras et al., 2024). Another study found that information from a food scanner app could encourage people to choose healthier products, although it did not always affect their actual purchases (Werle et al., 2024). These findings show that digital food systems can give consumers useful information while choosing products. Adding personalized information could make the system more useful by helping users focus on ingredients that matter to their own needs.

Barcode scanning can make checking packaged food easier since users do not have to search for every product manually. It was used by 13 out of 46 food purchasing apps to identify food products (Benthem de Grave et al., 2024). Barcode scanner applications can also provide information about food quality and its components to give consumers another way to learn more about the products they buy (Hanras et al., 2024). This can save time compared with manually searching for a product and its ingredients online. Barcode scanning can be used in the proposed system to give users quick access to ingredient information and help them make more informed choices when buying packaged food.

One such approach is to leverage digital technology in order to display food labels on user devices (e.g., laptops, smartphones) rather than requiring physical changes on product packages (Fuchs et al., 2022). Traceability systems are explored to track product information and location in the distribution process. Recently, one of the most popular traceability methods is two dimensional codes such as barcodes and QR (quick response) codes (Park et al., 2024). These

technologies provide ways to access food product information digitally without relying solely on the information printed on physical packages.

Although food labels provide important information, existing approaches do not always address consumers' practical information needs. Current food-information resources may have incomplete product coverage, limited explanations of technical ingredients, and insufficient representation of locally produced or small-business products (Inguglia et al., 2023; Wibowo & Hasnda, 2023). Consumer understanding also differs according to language, education, food literacy, product availability, and national labeling practices (Elrahi et al., 2023; Islam et al., 2024). In addition, digital food-information tools may face limitations involving database reliability, personalization, digital literacy, internet access, and user trust (Krish, 2025).

These limitations indicate a need for a localized and user-centered system that can connect a product with clear, personalized ingredient information. The proposed system addresses this gap by proposing barcode-based product retrieval, simplified ingredient explanations, and information tailored to users' dietary preferences or restrictions. The system should be evaluated in terms of database accuracy, local product coverage, usability, personalization, and usefulness for identifying ingredients that may affect consumer health or dietary decisions (Park et al., 2024).

A. Problem Being Addressed

Statement of the Problem

Consumers may struggle to understand ingredient labels and identify allergens because of dense text and technical terms. IngreGuide addresses this

problem by providing quick, simplified, and personalized ingredient information through barcode scanning.

General Question:

1. Can IngreGuide scan barcodes and provide a personalized food ingredients information system in response?

Specific Questions:

1. How effective is IngreGuide in providing personalized ingredient information to users in terms of:
    1.1. Barcode scanning accuracy;
    1.2. Task completion rate; and
    1.3. Information retrieval time?

B. Engineering Goals

This study aims to develop and evaluate IngreGuide, a web-based personalized food ingredient information system that uses camera-based barcode scanning to provide simplified ingredient information and identify ingredients relevant to users' dietary restrictions and preferences. Its engineering goal is to build a user-friendly platform where users can scan food product barcodes then navigate the system easily and complete their intended tasks efficiently.

C. Methodology

C1. Procedures

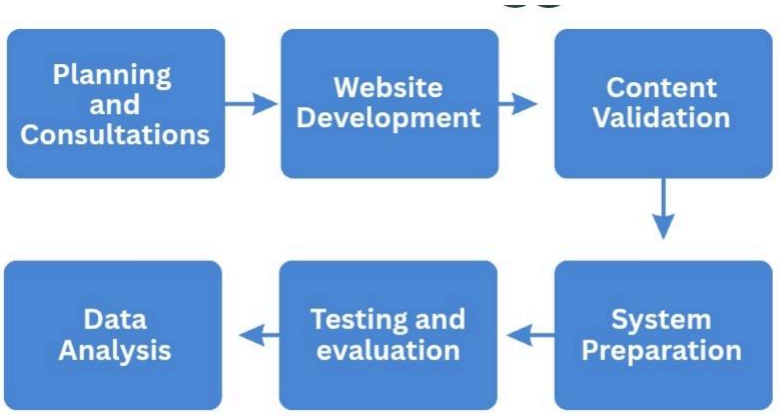


Figure 1. Flowchart of Procedures

C1.1. Research Design

This study utilizes a quantitative technical and evaluative research design. Such a design is suitable for systematically measuring, analyzing, and quantifying the functionality and operational performance of the IngreGuide web system across objective parameters, namely barcode scanning accuracy, task completion rate, and information retrieval time. Instead of recruiting general consumer end-users, system evaluation relies on controlled testing trials and expert verification of ingredient data performed by a purposely selected registered nutritionist-dietitian (RND) to guarantee data accuracy and user safety.

C1.2. Planning and Consultations

The researchers will first organise the research plan to identify the requirements and essential features of IngreGuide. With the consultations from the chosen qualified scientists, the researchers will work together with the programmer regarding the website's design, functionality, and technical requirements. Along with the nutritionist, where researchers will also work together with them regarding the accuracy and simplification of good ingredient information. Considering the revisions

and updates, the researchers are primarily responsible for relying necessary information on the research study towards the chosen qualified scientists for their professional guidance.

C1.3. Website Development

The programmer will mainly develop IngreGuide as a functional, web-based prototype by integrating appropriate frameworks, and a database to support its intended features. The use of a device's camera for barcode-based scanning softwares were observed and supported by de Grave et al. (2024), who then found that it is a feature used to identify products and display nutritional information. Meanwhile, the researchers will assist in the development by providing the necessary food product information for the database, and organising the user interface.



Figure 2. Login System User Interface Concept

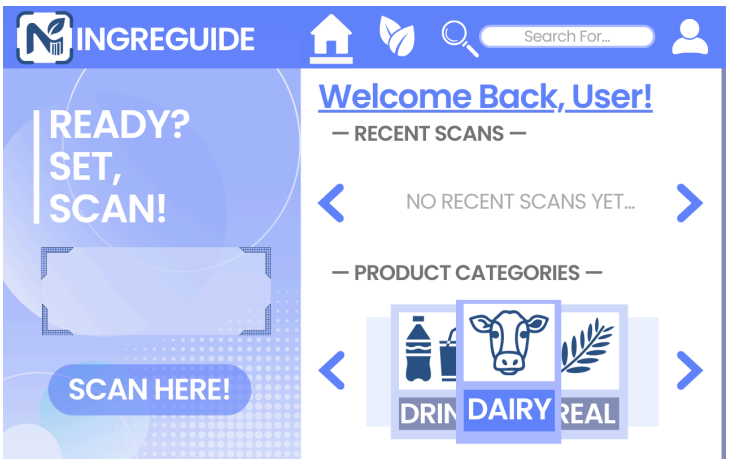


Figure 3. Home Page User Interface Concept

C1.4. Content Validation

The researchers will submit the chosen food products and their corresponding simplified ingredient explanations to a registered nutritionist and dietitian (RND) for a proper review and validation. Given the information, the nutritionist will assess the accuracy, clarity, and appropriateness of each explanation to ensure that the information presented by IngreGuide is consistent with the matching product barcode (Indeed Editorial Team, 2025). Based on the nutritionist's feedback, it will then determine if the adjusted information is suitable for inclusion in the website.

C1.5. System Preparation

After the content to be added is validated, the researchers will organise and coordinate with the programmer to ensure that the validated information is appropriately added in the website's database. Additionally, the researchers will then select the chosen food products to be used for testing and data-recording sheets needed to document the system's performance for the testing and evaluation phase.

C1.6. Testing and Evaluation

The researchers will test and evaluate IngreGuide based on the selected parameters to assess its performance in executing its intended features. The succeeding section on data analysis, will present each parameter, along with its corresponding testing procedure and formula, which will be used to handle the data received during the testing. Both the programmer and researchers must be present to check if the website works.

C2. Risk and Safety

To ensure the safety and responsible use of IngreGuide, the researchers will address potential risks related to improper and inaccurate displays of the ingredient information, and protection of users' personal data. To minimise the risk of misleading users, the food product information and simplified ingredient explanations must first be reviewed then verified by a registered nutritionist and dietitian (RND) before being incorporated into the website. Moreover, accurate ingredient information is important for individuals with dietary preferences or restrictions related to their health, such as allergies, as they need to identify and avoid ingredients that may trigger health-related reactions (U.S. Food and Drug Administration [FDA], 2025). Furthermore, the researchers will coordinate with the programmer to protect any personal information collected through IngreGuide, in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173). Considering the recommendations of the Web Content Accessibility Guidelines (WCAG) 2.2 (World Wide Web Consortium [W3C], 2024), the website will present information in a clear and accessible manner, which both the researchers and programmer must consider during the website's development. Since the study is a web-based system, the researchers will prioritise

the safety considerations relating to the information presented and the users' personal data throughout the study.

C3. Data Analysis

In this research, the researchers will focus on three main performance metrics to assess the efficiency of the IngreGuide system: Barcode Scanning Accuracy, Task Completion Rate, and Information Retrieval Time.

Barcode Scanning Accuracy

To evaluate barcode scanning accuracy, the researchers will test IngreGuide with the selected packaged food products. Each scanning trial will be checked to determine whether the scanned barcode leads to the correct corresponding food product. The number of correct product matches will be divided by the total number of scanning trials and multiplied by 100 to obtain the percentage of correct matches. This calculation adapts the match-rate approach described by Lai and Wong (2023), following the formula:

$$\text{Scanning Accuracy } (\%) = \frac{\text{Number of Correct Product Matches}}{\text{Total Number of Scanning Trials}} \times 100$$

Task Completion Rate

To determine the task completion rate, the researchers will assign predefined tasks that represent the intended functions of IngreGuide. Each attempt will be recorded as successful or unsuccessful according to the established completion criteria. The number of successfully completed tasks will be divided by the total

number of attempted tasks and multiplied by 100. This formula is based on the completion-rate measurement used by Park et al. (2024):

$$\text{Task Completion Rate (\%)} = \frac{\text{Number of Successfully Completed Tasks}}{\text{Total Number of Attempted Tasks}}$$

Information Retrieval Time

To measure the information retrieval time, the researchers will make use of IngreGuide by recording the time required for the website to display the corresponding food ingredient information after a valid barcode scan is submitted. The mean retrieval time will be calculated by summing the recorded times from all valid trials and dividing the total by the number of valid trials. This approach is consistent with the use of average retrieval time in information retrieval research (Li, 2026), while Park et al. (2024) supports measuring task completion time in a food-information system, applying the formula:

$$\text{MRT} = \frac{1}{n} \sum_{i=1}^{n} t_i$$

Where:

ti = information retrieval time recorded for the i-th valid trial

n = total number of valid trials

MRT = mean information retrieval time

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